

Litesh Perumalla

972.626.4159 | liteshperumalla@gmail.com | <https://www.linkedin.com/in/perumalla-litesh/> | <https://github.com/liteshperumalla>

PROFESSIONAL SUMMARY

Motivated Data Science graduate student with hands-on experience in AI, machine learning, and intelligent agent design. Skilled in building and deploying real-world applications across education, healthcare, and fraud detection. Proficient in NLP, deep learning, and Generative AI tools such as LangChain, LlamaIndex, and LangGraph Known for blending strong technical skills with a problem-solving mindset and a collaborative approach. Eager to contribute to impactful AI-driven solutions through innovation, experimentation, and responsible development.

EDUCATION

University of North Texas | Denton, TX - Master of Science – Data Science

Expected: Dec 2025

TECHNICAL SKILLS

Natural Language Processing: Text parsing, Sentiment Analysis, Transformers, BERT.

Deep Learning: Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), TensorFlow, PyTorch, Keras.

Generative AI: RAG (Retrieval-Augmented Generation), AI Agents, Vector Database, Prompt Engineering, Embeddings.

Big Data and Cloud Technologies: Amazon Web Services, Google Cloud Platform.

LLM: OpenAI, Ollama, Llama Index, Lang Chain, Lang Graph, Fine Tuning.

Database Management: PostgreSQL, MongoDB, Chroma db, FAISS.

Programming Languages: Python, R, SQL, C#, JavaScript, TypeScript.

Version Control & DevOps: Git, APIs

Certifications: Machine Learning Specialization- Coursera.

EMPLOYMENT EXPERIENCE

Discovery Park Library Services Academic Assistant | Denton, Texas

August 2024 - Present

- Consistently provide timely and accurate in-person, phone, and virtual assistance to over **100 students** and faculty each week, improving access to library services and resources.
- Work collaboratively with faculty to identify relevant academic materials, contributing to more effective teaching outcomes and impactful research efforts.

PROJECT EXPERIENCE

Smart Tutor AI – Personalized Teaching Support

Jan 2025 - Present

- Led the design and development of an AI-powered tutoring system that delivers course-specific responses using validated academic materials, enhancing learning Faithfulness and relevance by **semantic chunking**, Hybrid Retrievers.
- Built an efficient retrieval pipeline using metadata tagging and vector search to improve content discoverability and user interaction.
- **Tools used:** RAG, Ollama, Llama Index, Chroma db, Sentence-transformers, Vector Embeddings, BERT, Stream lit.

Skin Cancer Detection Using a Convolutional Neural Network

Nov 2024

- Implemented a CNN model to classify skin lesions from over 10,000 medical images, achieving 85% accuracy in early detection.
- Applied effective tuning and data augmentation techniques to enhance the model's generalization and robustness.
- **Tools used:** Python, Matplotlib, Sci-kit learn, NumPy, pandas, TensorFlow, CNN.

Fraud Detection using Machine Learning.

Oct 2024

- Developed a machine learning model capable of identifying fraudulent patterns within a dataset of **1 million+** transactions with high precision.
- Engineered features and trained multiple models, selecting the most accurate model based on performance metrics.
- Developed a web application to integrate the fraud detection model for real-time prediction and accessibility.
- Optimized model through **hyper parameter tuning**, achieving high accuracy and reliability in fraud detection.
- **Tools used:** Python, Sci-kit learn, Stream lit, Machine Learning Libraries.

Liver Disorder Using Machine Learning.

May 2024

- Built an ANN model that achieved over **90% accuracy** in diagnosing liver disorders using structured clinical data.
- Improved model performance through careful feature selection and normalization techniques.
- **Tools used:** Python, Matplotlib, Sci-kit learn, NumPy, pandas, TensorFlow, MLP Classifier, ANN.